

Python in Finance Finance in Python

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"Price is what you pay. Value is what you get."

— Warren Buffett

About Me

- Pure mathematician by training
- Research: Functional analysis, operator algebras, topology

Industry Experience

- Worked for major financial institutions (Swiss, US & UK)
- Developed mid-size legacy Python libraries
- Conducted numerous technical job interviews

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Contact

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MS Teams: (preferred for quick IMs)

Office Hours

Fridays 12:00–14:00 (in person)
or by appointment (online)

Lecture Schedule

- No lectures on Wednesday, October 23, 2024
- Replacement slot: TBA (remote)
- Emergency remote sessions possible—check MS Teams

Communication

- **Primary platform:** MS Teams
- Short IMs are fine, but please don't overuse
- **Always announce yourself** before office visits

Final Exam

- Written form, 90 minutes
- Focus: mathematics and financial theory (not coding)
- Mock exam paper will be provided
- All tutorial grades give a pass

Core Textbooks

- J.C. Hull, *Options, Futures and Other Derivatives*, Pearson, 2009
- B. Lubanovic, *Introducing Python*, 2nd Ed., O'Reilly, 2019
- Y. Hilpisch, *Python for Finance*, O'Reilly, 2014

Online Practice

- [hackerrank.com](https://www.hackerrank.com) — Python coding challenges
- projecteuler.net — Mathematical programming problems
- Stack Exchange: stackoverflow.com, quant.stackexchange.com

LLMs as Learning Tools

- Use ChatGPT, Claude, or Gemini as your personal Python syntax tutor!
- Great for explaining error messages, debugging, code review
- **Caveat:** Always verify answers—LLMs can hallucinate
- Not a substitute for understanding the fundamentals

Lectures cover two main areas:

1. **Finance:** Various instruments, their modelling, real-life examples
2. **Mathematics:** Rigorous proofs of all theorems needed for models

Tutorials are coding-oriented:

- Implementation of algorithms and models
- Weekly/biweekly take-home problems (submit via GitHub)
- Final project: Build a model, explain it, present your code

Mathematical prerequisites: Basic probability, calculus, linear algebra.

Programming prerequisites: None (we start from scratch)!

Part I

Financial Instruments

Equity: The Big Picture

Definition: Equity represents ownership in a company.

Two key valuations:

Book value: Assets minus liabilities (accounting value)

Market value: What investors are willing to pay (market cap)

Example (Tesla, early 2020):

- Book value: $\approx$ \$6 billion
- Market cap: $\approx$ \$100 billion

Why the huge difference? **Growth expectations!**

Key insight: Markets price the *future*, not just today's balance sheet.

Stock = a unit of equity

If a company has N shares outstanding and you own k shares:

$$\text{Your ownership fraction} = \frac{k}{N}$$

Example (Apple, 2023):

- Shares outstanding: ≈ 15.7 billion
- Market cap: $\approx \$2.8$ trillion
- Price per share: $\approx \$178$

Why do companies issue stock?

- Raise capital without incurring debt
- Founders can monetise their work
- Employees can be compensated with stock options

Public vs. Private Companies

Feature	Public	Private
Shares traded on exchange	Yes	No
Price discovery	Continuous	Infrequent
Regulatory requirements	Extensive (SEC, etc.)	Minimal
Reporting requirements	Quarterly/annual	Limited
Liquidity	High	Low
Example	Apple, Google	SpaceX, Stripe

Going public: Initial Public Offering (IPO)

Going private: Buyout (e.g., Elon Musk & Twitter, 2022)

Forward Contracts: Locking in Future Prices

Definition: A forward contract is an agreement to:

- Buy or sell an asset
- At a specified future date
- For a price agreed today

Key terminology:

Spot price (S_0): Current market price

Forward price (F): Agreed price for future delivery

Delivery date (T): When the exchange happens

Long position: Agrees to *buy* at F

Short position: Agrees to *sell* at F

Important: No money changes hands at $t = 0$ (unlike options)!

Scenario (March):

- A Polish farmer will harvest 100 tonnes of wheat in September
- Current wheat price: 1,200 PLN/tonne
- Farmer worries prices might fall by harvest time

Solution: Enter a forward contract

- Forward price: $F = 1,250$ PLN/tonne
- Farmer takes the **short** position (agrees to sell)
- Counterparty (e.g., a bakery) takes the **long** position

What happens in September?

Scenario	Spot price S_T	Farmer's payoff
Prices fell	1,100 PLN	+15,000 PLN
Prices rose	1,400 PLN	−15,000 PLN

Payoff per tonne:

$$\text{Payoff} = F - S_T = 1,250 - S_T$$

Forward Pricing: No-Arbitrage Argument

Question: What should the forward price F be?

Setup:

- Asset with spot price S_0
- Risk-free interest rate r (continuously compounded)
- Forward delivery at time T

Claim: The fair forward price is $F = S_0 e^{rT}$

Proof strategy: Show that any other price creates arbitrage.

Arbitrage strategy:

1. At $t = 0$: Borrow S_0 at rate r , buy the asset, enter short forward
2. At $t = T$: Deliver asset under forward, receive F , repay loan $S_0 e^{rT}$

Cash flows:

Time	Action	Cash
$t = 0$	Borrow S_0	$+S_0$
$t = 0$	Buy asset	$-S_0$
$t = 0$	Enter short forward	0
$t = T$	Deliver asset, receive F	$+F$
$t = T$	Repay loan	$-S_0 e^{rT}$
Net		$F - S_0 e^{rT} > 0$

Risk-free profit $\Rightarrow$ arbitrage $\Rightarrow$ market will correct!

Arbitrage strategy:

1. At $t = 0$: Short-sell the asset (receive S_0), invest at rate r , enter long forward
2. At $t = T$: Receive $S_0 e^{rT}$ from investment, pay F under forward, return asset

Cash flows:

Time	Action	Cash
$t = 0$	Short-sell asset	$+S_0$
$t = 0$	Invest at rate r	$-S_0$
$t = 0$	Enter long forward	0
$t = T$	Receive investment	$+S_0 e^{rT}$
$t = T$	Pay F , receive asset	$-F$
$t = T$	Return asset to lender	0
Net		$S_0 e^{rT} - F > 0$

Conclusion: $F = S_0 e^{rT}$ is the only arbitrage-free price. ■

Hidden assumptions behind F

We implicitly assumed:

- No transaction costs, no bid/ask spreads, continuous borrowing/lending at the same r .
- The asset can be short-sold and delivered at T .
- No interim cashflows from holding the asset (no dividends, coupons, storage costs, etc.).

Common generalisations:

- Continuous dividend yield q : $F_0 = S_0 e^{(r-q)T}$.
- Cost-of-carry (storage u , convenience yield y): $F_0 = S_0 e^{(r+u-y)T}$.

Feature	Forward	Future
Trading venue	OTC (private)	Exchange
Standardisation	Customisable	Standardised
Counterparty risk	Direct exposure	Cleared (low risk)
Settlement	At maturity only	Daily (mark-to-market)
Liquidity	Lower	Higher
Margin requirements	Negotiated	Required by exchange

Daily settlement (mark-to-market):

At end of each day, gains/losses are transferred between counterparties.

This reduces counterparty risk but requires liquidity management.

Options: The Right, Not the Obligation

Definition: An option gives the holder the *right* (but not obligation) to:

- Buy (**call**) or sell (**put**) an asset
- At a specified price (strike)
- On or before a specified date (maturity)

Key difference from forwards: Options are *asymmetric*!

Exercise styles:

- **European:** Can only exercise at maturity T
- **American:** Can exercise any time up to T

Terminology:

- **Strike price (K):** The pre-agreed price
- **Premium:** Price paid upfront for the option
- **Expiration/Maturity (T):** When the option expires

Option Payoffs at Maturity

Let S_T be the stock price at maturity.

Call option payoff:

$$\text{Payoff}_{\text{call}} = \max(S_T - K, 0) = (S_T - K)^+$$

You exercise only if $S_T > K$ (stock is worth more than strike).

Put option payoff:

$$\text{Payoff}_{\text{put}} = \max(K - S_T, 0) = (K - S_T)^+$$

You exercise only if $S_T < K$ (stock is worth less than strike).

Key asymmetry:

- Maximum loss for option buyer: the premium paid
- Maximum gain for option buyer: unlimited (call) or K (put)

Scenario (June):

- Google stock trades at \$561.32 (bid) / \$561.51 (offer)
- You buy 1 call option contract (= 100 shares)
- Strike price: $K = \$580$
- Expiration: December
- Premium: \$35.30 per share

Your initial cost: $100 \times \$35.30 = \$3,530$

What happens in December?

Options Example: Possible Outcomes

Scenario A: Google closes at \$550 in December

- $S_T = \$550 < K = \580
- Option expires worthless (you don't exercise)
- **Your loss:** \$3,530 (the premium)

Scenario B: Google closes at \$600 in December

- $S_T = \$600 > K = \580
- You exercise: buy 100 shares at \$580, worth \$600 each
- Gross profit: $100 \times (\$600 - \$580) = \$2,000$
- **Net profit:** $\$2,000 - \$3,530 = -\$1,530$ (still a loss!)

Scenario C: Google closes at \$650 in December

- Gross profit: $100 \times (\$650 - \$580) = \$7,000$
- **Net profit:** $\$7,000 - \$3,530 = \$3,470$

Break-even: $S_T = K + \text{premium} = \$580 + \$35.30 = \615.30

Bonds: Lending to Companies and Governments

Definition: A bond is a debt instrument where:

- The **issuer** (borrower) promises to pay
- The **bondholder** (lender) receives periodic interest + principal

Key terms:

Face value (par): Amount repaid at maturity (e.g., \$1,000)

Coupon rate: Annual interest as % of face value

Coupon payment: Actual cash received periodically

Maturity date: When principal is repaid

Yield: Effective return based on current price

Example: A \$1,000 bond with 5% coupon pays \$50/year in interest.

Key principle: A bond's price is the present value of all future cash flows.

Future cash flows:

- Coupon payments: C each period for N periods
- Face value: M at maturity

Present value formula:

$$P = \sum_{n=1}^N \frac{C}{(1+i)^n} + \frac{M}{(1+i)^N}$$

where i is the market interest rate (yield).

Closed form:

$$P = C \cdot \frac{1 - (1+i)^{-N}}{i} + M \cdot (1+i)^{-N}$$

Critical relationship: Bond prices and yields move in **opposite** directions!

Why?

- If market rates rise, new bonds offer higher coupons
- Your old bond (with lower coupon) becomes less attractive
- Its price must fall to offer competitive yield

Numerical example:

- Bond: \$1,000 face, 5% coupon, 10 years to maturity
- If market rate = 5%: Price $\approx$ \$1,000 (at par)
- If market rate = 6%: Price $\approx$ \$926 (discount)
- If market rate = 4%: Price $\approx$ \$1,081 (premium)

This is why bond portfolios lose value when central banks raise rates!

Bond Pricing: Clean vs. Dirty Price

Problem: Bonds pay coupons periodically (e.g., semi-annually).

What if you buy between coupon dates?

Accrued interest: Interest earned since last coupon payment.

Two price conventions:

- **Clean price:** Quoted price, excludes accrued interest
- **Dirty price:** Actual price paid = Clean + Accrued interest

Example:

- Bond pays \$50 coupon every 6 months
- You buy 3 months after last coupon
- Accrued interest: $\$50 \times \frac{3}{6} = \25
- If clean price is \$980, you pay \$1,005 (dirty price)

Part II

Risk Measures

Key questions for any portfolio:

- How much could I lose?
- How likely is a catastrophic loss?
- How much capital should I hold as a buffer?

Stakeholders who care:

- **Regulators:** Ensure banks can survive crises
- **Investors:** Understand portfolio risk
- **Risk managers:** Set limits and monitor exposures

The challenge: How do we quantify “risk” with a single number?

We need a function $\rho : \mathcal{X} \rightarrow \mathbb{R}$ that maps random losses to capital requirements.

Setting: Let $\mathcal{X}$ be a space of random variables representing portfolio losses over some horizon.

Definition: A *risk measure* is a function

$$\rho : \mathcal{X} \rightarrow \mathbb{R}$$

where $\rho(X)$ represents the capital needed to make position X “acceptable.”

Convention:

- $X > 0$ means a loss
- $X < 0$ means a profit
- Higher $\rho(X)$ = more capital required = riskier position

Question: What properties should a “good” risk measure have?

Definition (Artzner et al.

A risk measure ρ is *coherent* if it satisfies the four axioms below.

- 1. Monotonicity:** If $X \leq Y$ almost surely, then $\rho(X) \leq \rho(Y)$.
 - *Intuition:* Larger losses require more capital.
- 2. Translation invariance:** $\rho(X + c) = \rho(X) + c$ for any constant c .
 - *Intuition:* Adding c to losses increases capital need by exactly c .
- 3. Positive homogeneity:** $\rho(\lambda X) = \lambda \rho(X)$ for $\lambda > 0$.
 - *Intuition:* Doubling position size doubles the risk.
- 4. Subadditivity:** $\rho(X + Y) \leq \rho(X) + \rho(Y)$.
 - *Intuition:* Diversification should not increase risk.

Why Subadditivity Matters

Subadditivity: $\rho(X + Y) \leq \rho(X) + \rho(Y)$

Economic meaning:

- Merging two portfolios should not create *extra* risk
- Diversification benefits should be recognized
- A regulator can sum subsidiary risks to bound parent risk

Practical consequence:

If a bank has two trading desks with risks $\rho(X)$ and $\rho(Y)$:

- Total capital needed: at most $\rho(X) + \rho(Y)$
- No “hidden” risk created by combining them

Spoiler: Value at Risk (VaR) violates subadditivity in general!

Definition: Value at Risk

The Value at Risk at confidence level $\alpha \in (0, 1)$ is:

$$\text{VaR}_\alpha(X) = \inf\{m \in \mathbb{R} : P(X \leq m) \geq \alpha\} = F_X^{-1}(\alpha)$$

where F_X is the CDF of losses X .

Plain English:

“With probability α , losses will not exceed VaR_α .”

Typical values:

- $\alpha = 0.95$ (5% chance of exceeding VaR)
- $\alpha = 0.99$ (1% chance of exceeding VaR)

Example: If 1-day 99% VaR = \$10 million, then:

“We are 99% confident we won’t lose more than \$10M tomorrow.”

VaR: The Truck and Bridge Analogy

Imagine: A bridge with a posted weight limit of 10 tonnes.

Question: Is it safe for a fleet of trucks to cross?

VaR approach:

- Measure the weight of each truck
- 99% of trucks weigh $\leq$ 10 tonnes
- **Conclusion:** “99% VaR = 10 tonnes, bridge is fine!”

The problem:

- VaR says nothing about the 1% of trucks that exceed 10 tonnes
- One might weigh 11 tonnes (minor issue)
- Another might weigh 50 tonnes (bridge collapses!)

Lesson: VaR tells you the threshold, not what happens beyond it.

1. VaR is NOT subadditive in general!

Example: Two identical bonds, each defaults with probability 4%.

- Individual 95% VaR: 0 (each survives with 96% probability)
- Combined: 95% VaR could be positive (both defaulting is rare, but one defaulting is $\approx 8\%$)

$\Rightarrow$ VaR can penalize diversification!

2. VaR ignores tail severity

- Two portfolios can have the same VaR
- But very different losses when VaR is breached

3. Not coherent $\Rightarrow$ can lead to inconsistent decisions

Despite these flaws, VaR remains popular due to its simplicity and regulatory entrenchment (Basel accords).

VaR Calculation Methods: Overview

Three main approaches to compute VaR:

Method	Speed	Accuracy	Non-linear?
Parametric (Delta-Normal)	Fast	Low	Poor
Historical Simulation	Medium	Medium	Good
Monte Carlo	Slow	High	Good

Choice depends on:

- Portfolio complexity (linear vs. options-heavy)
- Available computational resources
- Required accuracy and regulatory context

We model P&L and then define loss.

Let ΔV be 1-day P&L (positive = profit). Assume

$$\Delta V \sim N(\mu_{\Delta V}, \sigma_{\Delta V}^2).$$

Define the *loss* $L := -\Delta V$ (positive = loss). Then

$$L \sim N(-\mu_{\Delta V}, \sigma_{\Delta V}^2).$$

VaR of loss

$$\text{VaR}_{\alpha}(L) = \inf\{m : P(L \leq m) \geq \alpha\} = -\mu_{\Delta V} + z_{\alpha}\sigma_{\Delta V}, \quad z_{\alpha} := \Phi^{-1}(\alpha).$$

Rule of thumb: for short horizons often $\mu_{\Delta V} \approx 0$, so $\text{VaR}_{0.99}(L) \approx 2.326 \sigma_{\Delta V}$.

Portfolio case: $\sigma_{\Delta V} = \sqrt{\mathbf{w}^T \mathbf{\Sigma} \mathbf{w}} \times V$ where $\mathbf{w}$ are weights and $\mathbf{\Sigma}$ is the return covariance matrix.

Setup:

- Portfolio value: \$10 million
- Two assets with weights: $w_1 = 0.6$, $w_2 = 0.4$
- Daily volatilities: $\sigma_1 = 2\%$, $\sigma_2 = 3\%$
- Correlation: $\rho_{12} = 0.4$

Step 1: Compute portfolio variance:

$$\begin{aligned}\sigma_V^2 &= w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2 \\ &= 0.36 \times 0.0004 + 0.16 \times 0.0009 + 2 \times 0.24 \times 0.4 \times 0.0006 = 0.000403\end{aligned}$$

$$\sigma_V = 2.01\%$$

Step 2: Compute 99% VaR:

$$\text{VaR}_{0.99} = 2.326 \times 0.0201 \times \$10\text{M} = \$467,500$$

The normality assumption is dangerous:

1. Fat tails in reality

- Market returns have excess kurtosis (“fat tails”)
- Extreme events occur more often than normal predicts
- 2008 financial crisis: 25+ standard deviation moves!

2. Non-linear instruments

- Options have non-linear payoffs (γ , convexity)
- Delta-normal ignores this entirely
- Severely underestimates risk for options portfolios

3. Time-varying volatility

- Volatility clusters (GARCH effects)
- Constant σ assumption fails during crises

Quote: “Parametric VaR is like checking your speedometer while driving off a cliff.”

Historical Simulation VaR (Quantile Fixed)

Idea: Use historical returns to generate an empirical loss distribution.

Algorithm (losses):

1. Collect n historical returns: $r_1, \dots, r_n$.
2. Compute P&L scenarios ΔV_i and losses $L_i := -\Delta V_i$.
3. Sort losses in increasing order: $L_{(1)} \leq \dots \leq L_{(n)}$.
4. The empirical VaR at level α is the empirical α -**quantile**:

$$\text{VaR}_\alpha \approx L_{(\lceil \alpha n \rceil)}.$$

Example: 99% VaR with $n = 500$

- Worst 1% tail = 5 largest losses: indices 496, ..., 500.
- $\text{VaR}_{0.99} \approx L_{(496)} =$ the *5th worst* loss.

Pros: No distribution assumptions, captures tails and non-linearities.

Cons: Past $\neq$ future, window choice, regime changes.

Idea: Simulate thousands of possible future scenarios.

Algorithm:

1. Specify stochastic model for risk factors (e.g., GBM, GARCH)
2. Generate N simulated paths (e.g., $N = 10,000$)
3. For each path, compute portfolio value at horizon
4. $\text{VaR}_\alpha = (1 - \alpha)$ -quantile of simulated P&L

Example: 99% VaR for options portfolio

- Simulate 10,000 stock price paths using GBM
- Price all options in each scenario (full revaluation)
- Sort P&L, take 100th worst loss

Advantages: Flexible models, handles any instrument, captures correlations and fat tails.

Disadvantages: Computationally expensive, model risk (garbage in = garbage out).

Computation for 1-day 99% VaR, \$100M equity portfolio:

Method	Time	Result	Comment
Parametric	<1 sec	\$2.5M	Underestimates
Historical (500d)	2 sec	\$3.1M	Reasonable
Monte Carlo (10k)	30 sec	\$3.2M	Most accurate

When to use what:

- **Parametric:** Quick estimates, simple portfolios, educational
- **Historical:** Daily reporting, regulatory compliance
- **Monte Carlo:** Complex derivatives, stress testing, model validation

Industry practice: Most banks use all three, compare results.

VaR Aggregation: Risk Factor vs. Portfolio Level

Risk factor level: Compute VaR for individual risk factors

- Interest rate VaR, FX VaR, equity VaR, credit VaR
- Useful for understanding risk drivers

Portfolio level: Aggregate to total VaR

- Must account for correlations between factors
- Can't simply add individual VaRs (ignores diversification)

Aggregation approaches:

1. **Full simulation:** Simulate all factors jointly (best but slow)
2. **Variance-covariance:** $\text{VaR}_{\text{total}}^2 = \sum_{i,j} \text{VaR}_i \cdot \text{VaR}_j \cdot \rho_{ij}$
3. **Square root of sum:** $\text{VaR}_{\text{total}} \approx \sqrt{\sum_i \text{VaR}_i^2}$ (assumes zero correlation)

Diversification benefit: $\text{VaR}_{\text{total}} < \sum_i \text{VaR}_i$ (if $\rho < 1$)

Definition (robust)

For $\alpha \in (0, 1)$ and integrable loss X , define

$$\text{ES}_\alpha(X) = \frac{1}{1 - \alpha} \int_\alpha^1 \text{VaR}_u(X) \, du.$$

If X has a **continuous distribution**, this coincides with the conditional tail mean:

$$\text{ES}_\alpha(X) = \mathbb{E}[X \mid X > \text{VaR}_\alpha(X)].$$

Why use the integral definition?

It behaves well for discrete losses and is the standard definition in risk theory and regulation.

Also known as: CVaR (Conditional VaR), AVaR (Average VaR), Expected Tail Loss (ETL).

Expected Shortfall: Continuous Case

For continuous distributions:

$$\text{ES}_\alpha(X) = \frac{1}{1 - \alpha} \int_{\text{VaR}_\alpha}^{\infty} x f_X(x) dx$$

where f_X is the PDF of X .

Intuition: ES averages VaR across all confidence levels above α .

Key property: $\text{ES}_\alpha \geq \text{VaR}_\alpha$ always (ES is at least as conservative).

Theorem: Expected Shortfall is a *coherent* risk measure.

Dual representation (finite L^1 version)

For $\alpha \in (0, 1)$ and integrable loss X ,

$$\text{ES}_\alpha(X) = \sup \left\{ E[XZ] : Z \geq 0, E[Z] = 1, Z \leq \frac{1}{1-\alpha} \text{ a.s.} \right\}.$$

Equivalently, $Z = dQ/dP$ for measures $Q \ll P$ with $0 \leq dQ/dP \leq 1/(1-\alpha)$.

Subadditivity: for any X, Y ,

$$\text{ES}_\alpha(X + Y) = \sup_Z E[(X + Y)Z] \leq \sup_Z E[XZ] + \sup_Z E[YZ] = \text{ES}_\alpha(X) + \text{ES}_\alpha(Y).$$



Loss distribution with VaR and ES marked

Consider a loss distribution with:

- 95% VaR = \$1 million
- 95% ES = \$2.5 million

Interpretation:

- 95% of the time, we lose $\leq$ \$1M (VaR tells us this)
- When we lose $>$ \$1M, average loss is \$2.5M (ES tells us this)

The truck analogy revisited:

- VaR: “99% of trucks are under 10 tonnes”
- ES: “The 1% over 10 tonnes average 25 tonnes”

ES captures **tail risk severity**, not just frequency!

Basel II (1996–2012):

- 10-day 99% VaR for market risk capital
- Multiplier: $k \geq 3$ (capital = $k \times \text{VaR}$)
- Backtesting: count VaR breaches, increase k if too many

Basel III / FRTB (2016–present):

- Shift from VaR to **Expected Shortfall** (97.5% ES)
- Rationale: ES is coherent, captures tail risk
- Liquidity horizons: different for asset classes (10–120 days)

Key Basel ES parameters:

- Confidence level: 97.5% (equivalent severity to 99% VaR under normality)
- Base horizon: 10 days
- Stressed ES: Use crisis period data

Note: Banks must compute both VaR and ES for regulatory reporting.

Property	VaR	ES
Easy to compute	✓	✓
Easy to interpret	✓	Moderate
Subadditive	✗	✓
Coherent	✗	✓
Captures tail risk	✗	✓
Regulatory standard	✓ (Basel II/III)	✓ (Basel III.1/FRTB)

Practical advice:

- Use VaR for quick communication and legacy compliance
- Use ES for internal risk management and portfolio optimization
- Always look at both, plus stress tests and scenario analysis

Part III

Probability Foundations

Question: What is $E[X \mid \mathcal{G}]$?

Intuition: It's your "best guess" of X given information $\mathcal{G}$.

Three key perspectives:

1. **Information:** $E[X|\mathcal{G}]$ uses only info in $\mathcal{G}$
2. **Averaging:** It's X averaged over outcomes indistinguishable given $\mathcal{G}$
3. **Projection:** In L^2 , it's the orthogonal projection of X onto $\mathcal{G}$ -measurable functions

Example: Roll a die, let X = number shown.

- $E[X] = 3.5$ (no information)
- $E[X \mid X \text{ is even}] = 4$ (partial information)
- $E[X \mid X] = X$ (full information)

Conditional Expectation: Formal Definition

Setting: Probability space $(\Omega, \mathcal{F}, P)$, sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$.

Definition: Conditional Expectation

For integrable X , the conditional expectation $E[X|\mathcal{G}]$ is the (a.s. unique) r.v. satisfying:

1. $E[X|\mathcal{G}]$ is $\mathcal{G}$ -measurable
2. For all $G \in \mathcal{G}$: $\int_G E[X|\mathcal{G}] dP = \int_G X dP$

Plain English:

- Condition 1: Result only uses information in $\mathcal{G}$
- Condition 2: Averages correctly over all $\mathcal{G}$ -events

Existence: Guaranteed by Radon-Nikodym theorem!

Absolute Continuity

Let μ and ν be measures on $(\Omega, \mathcal{F})$. We say ν is **absolutely continuous** with respect to μ , written $\nu \ll \mu$, if:

$$\mu(A) = 0 \implies \nu(A) = 0 \quad \text{for all } A \in \mathcal{F}$$

Intuition: ν “doesn’t see” sets that μ considers negligible.

Examples:

- If $\nu(A) = \int_A f d\mu$ for some $f \geq 0$, then $\nu \ll \mu$
- On $\mathbb{R}$: Lebesgue measure $\lambda \ll \lambda$, but $\delta_0 \not\ll \lambda$ (Dirac measure)
- If Q is risk-neutral and equivalent to P , then $Q \ll P$ and $P \ll Q$

Equivalent measures: $\mu \sim \nu$ iff $\mu \ll \nu$ and $\nu \ll \mu$, i.e., they have the same null sets.

Radon–Nikodym Theorem

Let $(\Omega, \mathcal{F})$ be a measurable space and let μ, ν be σ -finite measures with $\nu \ll \mu$. Then there exists a measurable function $f: \Omega \rightarrow [0, \infty)$ such that:

$$\nu(A) = \int_A f \, d\mu \quad \text{for all } A \in \mathcal{F}$$

Moreover, f is unique μ -a.e. We write $f = \frac{d\nu}{d\mu}$.

Assumptions explained:

- **σ -finite:** $\Omega = \bigcup_{n=1}^{\infty} A_n$ with $\mu(A_n) < \infty$. Ensures integrals are well-defined.
- **Absolute continuity:** Essential! Without $\nu \ll \mu$, no density exists.

Key application: If P and Q are equivalent probability measures, then:

$$E^Q[X] = E^P \left[X \cdot \frac{dQ}{dP} \right]$$

Proof of Radon–Nikodym (Schep's Approach, Clean Version)

Assume first $\mu(\Omega) < \infty$ and $\nu(\Omega) < \infty$ (reduce general case by σ -finiteness).

Define

$$\mathcal{G} = \left\{ g \geq 0 \text{ measurable} : \int_A g \, d\mu \leq \nu(A) \, \forall A \in \mathcal{F} \right\}.$$

Then $\mathcal{G} \neq \emptyset$ (take $g = 0$) and $\sup_{g \in \mathcal{G}} \int g \, d\mu \leq \nu(\Omega) < \infty$.

Choose $(g_n) \subset \mathcal{G}$ with $\int g_n \, d\mu \uparrow \sup_{g \in \mathcal{G}} \int g \, d\mu$ and set $f_n = \max(g_1, \dots, g_n)$. Then $f_n \in \mathcal{G}$ and $f_n \uparrow f$ pointwise.

By MCT, $\int f_n \, d\mu \uparrow \int f \, d\mu$, and one checks $f \in \mathcal{G}$, hence

$$\int f \, d\mu = \sup_{g \in \mathcal{G}} \int g \, d\mu.$$

Proof of Radon–Nikodym (Conclusion via Hahn Decomposition)

Define the finite measure

$$\rho(A) := \nu(A) - \int_A f \, d\mu \quad (A \in \mathcal{F}).$$

Since $f \in \mathcal{G}$, we have $\rho(A) \geq 0$ for all A . We show $\rho \equiv 0$.

Assume for contradiction that $\rho(B) > 0$ for some B . Then $\mu(B) > 0$ (because $\rho \ll \mu$ follows from $\nu \ll \mu$). Set $\varepsilon := \rho(B)/(2\mu(B)) > 0$ and consider the signed measure on $\mathcal{F}$:

$$\eta(A) := \rho(A \cap B) - \varepsilon \mu(A \cap B).$$

Then $\eta(B) = \rho(B) - \varepsilon \mu(B) = \rho(B)/2 > 0$.

By the Hahn decomposition theorem applied to η , there exists a measurable $C \subseteq B$ (a “positive set”) such that

$$\eta(A) \geq 0 \quad \text{for all } A \subseteq C, \quad \text{i.e.} \quad \rho(A) \geq \varepsilon \mu(A) \quad \text{for all } A \subseteq C.$$

Define $\tilde{g} := f + \varepsilon 1_C$. For any A ,

$$\int_A \tilde{g} \, d\mu = \int_A f \, d\mu + \varepsilon \mu(A \cap C) \leq \int_A f \, d\mu + \rho(A \cap C) \leq \nu(A),$$

so $\tilde{g} \in \mathcal{G}$. But $\int \tilde{g} \, d\mu = \int f \, d\mu + \varepsilon \mu(C) > \int f \, d\mu$, contradicting maximality of f . Hence $\rho \equiv 0$, so $\nu(A) = \int_A f \, d\mu$ for all A . ■

Construction of Conditional Expectation via Radon–Nikodym

Constructing $E[X|\mathcal{G}]$:

Step 1: For $X \geq 0$, define measure ν on $(\Omega, \mathcal{G})$ by: $\nu(G) = \int_G X dP = E[X \cdot 1_G]$

Step 2: Check $\nu \ll P|_{\mathcal{G}}$ (if $P(G) = 0$ then $\nu(G) = 0$).

Step 3: By Radon–Nikodym, $\exists \mathcal{G}$ -measurable Y with: $\nu(G) = \int_G Y dP$ for all $G \in \mathcal{G}$

Conclusion: Set $E[X|\mathcal{G}] := Y$. ■

General X

For signed X , write $X = X^+ - X^-$ and set $E[X|\mathcal{G}] = E[X^+|\mathcal{G}] - E[X^-|\mathcal{G}]$.

Conditional Expectation: Basic Properties

Let X, Y be integrable, $\mathcal{H} \subseteq \mathcal{G} \subseteq \mathcal{F}$.

1. **Linearity:** $E[aX + bY|\mathcal{G}] = aE[X|\mathcal{G}] + bE[Y|\mathcal{G}]$

2. **Tower property:** $E[E[X|\mathcal{G}]|\mathcal{H}] = E[X|\mathcal{H}]$

- *Intuition:* Coarsening information twice gives coarser result

3. **Taking out what is known:** If Y is $\mathcal{G}$ -measurable:

$$E[XY|\mathcal{G}] = Y \cdot E[X|\mathcal{G}]$$

4. **Independence:** If $X \perp \mathcal{G}$, then $E[X|\mathcal{G}] = E[X]$

5. **Trivial σ -algebra:** $E[X|\{\emptyset, \Omega\}] = E[X]$

6. **Full information:** $E[X|\sigma(X)] = X$

Conditional Expectation: More Properties

7. Monotonicity: If $X \leq Y$ a.s., then $E[X|\mathcal{G}] \leq E[Y|\mathcal{G}]$ a.s.

8. Jensen's inequality: If φ is convex:

$$\varphi(E[X|\mathcal{G}]) \leq E[\varphi(X)|\mathcal{G}]$$

9. Conditional Fatou: $E[\liminf X_n|\mathcal{G}] \leq \liminf E[X_n|\mathcal{G}]$

10. Dominated convergence: If $X_n \rightarrow X$ and $|X_n| \leq Y$ integrable:

$$E[X_n|\mathcal{G}] \rightarrow E[X|\mathcal{G}] \quad \text{a.s. and in } L^1$$

Proof technique: Most follow from the defining integral property plus corresponding unconditional results.

Proof of the Tower Property

Claim: $E[E[X|\mathcal{G}]|\mathcal{H}] = E[X|\mathcal{H}]$ for $\mathcal{H} \subseteq \mathcal{G}$.

Proof:

Let $Y = E[X|\mathcal{G}]$. We need to show $E[Y|\mathcal{H}] = E[X|\mathcal{H}]$.

Check the two defining conditions:

(1) $\mathcal{H}$ -measurability: $E[X|\mathcal{H}]$ is $\mathcal{H}$ -measurable by definition. ✓

(2) Integral condition: For any $H \in \mathcal{H}$:

Since $\mathcal{H} \subseteq \mathcal{G}$, we have $H \in \mathcal{G}$, so:

$$\int_H Y \, dP = \int_H E[X|\mathcal{G}] \, dP = \int_H X \, dP$$

by the defining property of $E[X|\mathcal{G}]$.

But also $\int_H E[X|\mathcal{H}] \, dP = \int_H X \, dP$.

So $\int_H Y \, dP = \int_H E[X|\mathcal{H}] \, dP$ for all $H \in \mathcal{H}$.

By uniqueness, $E[Y|\mathcal{H}] = E[X|\mathcal{H}]$. ■

Characteristic Functions: Definition

Definition: The characteristic function of random variable X is:

$$\varphi_X(t) = E[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} dF_X(x)$$

Why well-defined? $|e^{itX}| = 1$, so the integral always converges.

Relation to Fourier transform: If X has density f :

$$\varphi_X(t) = \int_{-\infty}^{\infty} e^{itx} f(x) dx = \widehat{f}(-t)$$

Key property: Characteristic function *uniquely determines* distribution.

Why useful?

- Converts convolutions (sums of r.v.s) to products
- Works even when moments don't exist
- Foundation for CLT proofs

Characteristic Functions: Examples

1. **Bernoulli(p):** $X \in \{0, 1\}$, $P(X = 1) = p$

$$\varphi_X(t) = (1 - p) + pe^{it} = 1 - p + pe^{it}$$

2. **Normal(μ, σ^2):**

$$\varphi_X(t) = \exp\left(i\mu t - \frac{\sigma^2 t^2}{2}\right)$$

3. **Poisson(λ):**

$$\varphi_X(t) = \exp(\lambda(e^{it} - 1))$$

4. **Discrete with values $\{2, 3, 7\}$, probs $\{0.2, 0.5, 0.3\}$:**

$$\varphi_X(t) = 0.2e^{2it} + 0.5e^{3it} + 0.3e^{7it}$$

Key formula for sums: If X, Y independent: $\varphi_{X+Y}(t) = \varphi_X(t) \cdot \varphi_Y(t)$

Lévy's Continuity Theorem

Convergence in distribution: $X_n \xrightarrow{d} X$ means:

$$F_{X_n}(x) \rightarrow F_X(x) \quad \text{at all continuity points of } F_X$$

Theorem (Lévy's Continuity Theorem)

Let (X_n) be a sequence of random variables with characteristic functions (φ_n) .

(a) If $X_n \xrightarrow{d} X$, then $\varphi_n(t) \rightarrow \varphi_X(t)$ for all t .

(b) If $\varphi_n(t) \rightarrow \varphi(t)$ for all t , and φ is continuous at 0, then φ is a characteristic function and $X_n \xrightarrow{d} X$ where $\varphi = \varphi_X$.

Power of Lévy: Reduces convergence in distribution to pointwise convergence of functions!

Theorem (Central Limit Theorem)

Let $X_1, X_2, \dots$ be i.i.d. with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$.

Let $S_n = X_1 + \dots + X_n$. Then:

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1)$$

Plain English: Normalized sums of i.i.d. variables converge to normal.

Why remarkable?

- Works for *any* distribution with finite variance
- Explains ubiquity of normal distribution in nature
- Foundation for statistics and finance

Central Limit Theorem: Proof via Characteristic Functions (Rigorous)

Assume WLOG $\mu = 0$, $\sigma = 1$ and set $Z_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n X_k$.

Step 1 (factorisation):

$$\varphi_{Z_n}(t) = \mathbb{E}[e^{itZ_n}] = \left(\varphi_X(t/\sqrt{n})\right)^n.$$

Step 2 (second-order expansion at 0): since $\mathbb{E}[X] = 0$, $\mathbb{E}[X^2] = 1$ and $|e^{isX}| \leq 1$,

$$\varphi_X(s) = 1 - \frac{s^2}{2} + o(s^2) \quad (s \rightarrow 0).$$

Step 3 (take logs):

$$\log \varphi_X(t/\sqrt{n}) = \log\left(1 - \frac{t^2}{2n} + o(1/n)\right) = -\frac{t^2}{2n} + o(1/n).$$

Multiply by n :

$$n \log \varphi_X(t/\sqrt{n}) \rightarrow -\frac{t^2}{2}.$$

Step 4 (exponentiate):

$$\varphi_{Z_n}(t) = \exp\left(n \log \varphi_X(t/\sqrt{n})\right) \rightarrow e^{-t^2/2}.$$

By Lévy's theorem, $Z_n \xrightarrow{d} N(0, 1)$. ■

Donsker's Theorem: Random Walk to Brownian Motion

Setup: Let $X_1, X_2, \dots$ be i.i.d. with $E[X_i] = 0$, $\text{Var}(X_i) = 1$.

Define the random walk: $S_n = X_1 + \dots + X_n$

Rescaled process: For $t \in [0, 1]$, define:

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} S_{\lfloor nt \rfloor} + \frac{nt - \lfloor nt \rfloor}{\sqrt{n}} X_{\lfloor nt \rfloor + 1}$$

(linear interpolation of $S_k/\sqrt{n}$ at times k/n)

Theorem (Donsker)

$W^{(n)} \xrightarrow{d} W$ in $C[0, 1]$, where W is standard Brownian motion.

Plain English: Rescaled random walks converge to Brownian motion!

Two ingredients:

1. Finite-dimensional convergence:

For fixed $0 < t_1 < \dots < t_k \leq 1$:

$$(W^{(n)}(t_1), \dots, W^{(n)}(t_k)) \xrightarrow{d} (W(t_1), \dots, W(t_k))$$

Proof: Multivariate CLT. Increments are sums of i.i.d. variables.

2. **Tightness:** The sequence $(W^{(n)})$ is tight in $C[0, 1]$.

Proof: Use moment bounds on increments:

$$\mathbb{E}[|W^{(n)}(t) - W^{(n)}(s)|^4] \leq C|t - s|^2$$

This implies tightness by Kolmogorov's criterion.

Conclusion: By Prokhorov's theorem, fidi convergence + tightness $\Rightarrow$ weak convergence in $C[0, 1]$.

Why Donsker Matters for Finance

Discrete world: Stock prices move in discrete steps at discrete times.

Continuous world: Mathematical models use continuous processes.

Donsker's bridge:

CRR Model	$\xrightarrow{\text{Donsker}}$	Black-Scholes
$S_n = S_0 \prod_{i=1}^n (1 + R_i)$		$S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$
$R_i \in \{u - 1, d - 1\}$		$W_t = \text{Brownian motion}$

The parallel:

- CLT: Sum of i.i.d. $\rightarrow$ normal
- Donsker: Process of sums $\rightarrow$ Brownian motion
- CRR $\rightarrow$ Black-Scholes: Binomial tree $\rightarrow$ GBM

All three are instances of the same phenomenon: **universality of the Gaussian.**

Do Non-Standard Methods Prove New Theorems?

Short answer: NSA does not prove *more* theorems than standard mathematics, but it often proves them *better*.

Foundational fact

Non-standard analysis is a *conservative extension* of ZFC: any first-order statement about $\mathbb{R}$, probability, or analysis proved using NSA *can in principle* be translated into a standard proof.

So why use NSA? Because many classical results:

- were **first proved using NSA**,
- admit **much simpler or more conceptual NSA proofs**,
- or become **structurally transparent** only in the hyperfinite picture.

Canonical examples:

- Loeb measure theorem (hyperfinite $\Rightarrow$ standard probability)
- Anderson's hyperfinite construction of Brownian motion (1976)
- LLN, CLT, Donsker theorem via a single infinite sample
- Compactness and existence results via near-standardness

Takeaway: NSA does not change *what* is true—it changes *how clearly* we see why it is true.

NSA as a change of coordinates

Standard analysis:

$$\text{finite } n \longrightarrow \lim_{n \rightarrow \infty}$$

Non-standard analysis:

$$\text{one infinite } N \longrightarrow \text{standard part}$$

Practical moral: Many “limit theorems” are really *algebraic statements about hyperfinite objects* whose content becomes obvious once limits are removed.

Example (CLT):

- Standard: $\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$
- NSA: For infinite N , $\frac{S_N - N\mu}{\sigma\sqrt{N}} \approx Z$ where Z is standard normal

No limits, no subsequences, no tightness arguments—just *one* infinite sample.

Part IV

Efficient Markets & No-Arbitrage

Efficient Market Hypothesis (EMH)

Core idea (Fama, 1970): Asset prices fully reflect all available information.

Three forms of efficiency:

Weak form: Prices reflect all *past* price data

⇒ Technical analysis doesn't work

Semi-strong form: Prices reflect all *public* information

⇒ Fundamental analysis doesn't work

Strong form: Prices reflect *all* information (including private)

⇒ Even insider trading doesn't work

Implication: You can't consistently "beat the market" ⇒ buy index funds!

The No-Arbitrage Principle

Definition: An *arbitrage* is a trading strategy that:

1. Requires zero initial investment
2. Has no risk of loss
3. Has positive probability of profit

No-Arbitrage Principle: In well-functioning markets, arbitrage opportunities don't persist.

Why?

- If arbitrage exists, traders exploit it immediately
- Their trades push prices until arbitrage disappears
- Markets are in equilibrium when no arbitrage remains

Consequence: We can derive prices by assuming no arbitrage.

This is the foundation of **all** derivatives pricing!

Model:

- Time: $t = 0$ (now) and $t = 1$ (future)
- K tradeable assets: $A_1, \dots, A_K$
- Asset A_1 is *riskless*: $S_1^1(\omega) = e^r S_0^1$ for all scenarios
- Finite set of N market scenarios: $\Omega = \{\omega_1, \dots, \omega_N\}$

Notation:

- S_0^j : Price of asset j at $t = 0$
- $S_1^j(\omega_i)$: Price of asset j at $t = 1$ in scenario ω_i

Portfolio: $\theta = (\theta_1, \dots, \theta_K) \in \mathbb{R}^K$

Value at $t = 0$: $V_0(\theta) = \sum_{j=1}^K \theta_j S_0^j$

Value at $t = 1$: $V_1(\theta; \omega_i) = \sum_{j=1}^K \theta_j S_1^j(\omega_i)$

Risk-Neutral Measures: Discounted Prices

Fix a physical measure P with full support: $P(\omega_i) > 0$ for all i .

Let $B_0 = 1$ and $B_1 = e^r$ be the money-market account (riskless asset). Define **discounted prices**

$$\tilde{S}_0^j := \frac{S_0^j}{B_0} = S_0^j, \quad \tilde{S}_1^j(\omega) := \frac{S_1^j(\omega)}{B_1} = e^{-r} S_1^j(\omega).$$

Risk-neutral / EMM in 1 period

A probability measure Q on Ω is *risk-neutral* if:

$$Q(\omega_i) > 0 \quad \forall i \quad \text{and} \quad \tilde{S}_0^j = E^Q[\tilde{S}_1^j] \quad \text{for all assets } j.$$

Interpretation: discounted asset prices have zero “excess drift” under Q .

Theorem (Harrison & Pliska)

A finite-state single-period market is **arbitrage-free** if and only if there exists a **risk-neutral probability measure**.

Two directions to prove:

($\Leftarrow$) Risk-neutral measure exists $\Rightarrow$ No arbitrage

(Relatively easy—we'll do this first)

($\Rightarrow$) No arbitrage $\Rightarrow$ Risk-neutral measure exists

(Harder—requires hyperplane separation theorem)

In finite-state markets the “hard” direction follows from a separating hyperplane theorem (or Farkas' lemma / linear programming duality).

FTAP Proof: Risk-Neutral $\Rightarrow$ No Arbitrage

Setup: Assume risk-neutral measure Q exists.

Goal: Show no arbitrage exists.

Proof:

Let θ be any portfolio. Its value at $t = 0$ is:

$$\begin{aligned} V_0(\theta) &= \sum_{j=1}^K \theta_j S_0^j = \sum_{j=1}^K \theta_j \cdot e^{-r} E^Q[S_1^j] \\ &= e^{-r} E^Q \left[\sum_{j=1}^K \theta_j S_1^j(\cdot) \right] = e^{-r} E^Q[V_1(\theta)] \end{aligned}$$

Now suppose θ is an arbitrage: $V_0(\theta) \leq 0$ and $V_1(\theta; \omega_i) \geq 0$ for all i , with strict inequality somewhere.

Then $E^Q[V_1(\theta)] > 0$ (since $Q(\omega_i) > 0$ for all i).

But $V_0(\theta) = e^{-r} E^Q[V_1(\theta)] > 0$. Contradiction! ■

Let $\tilde{S}_1^j \in \mathbb{R}^N$ be the vector of discounted payoffs:

$$\tilde{S}_1^j := (\tilde{S}_1^j(\omega_1), \dots, \tilde{S}_1^j(\omega_N))^\top.$$

For a portfolio $\theta \in \mathbb{R}^K$, the discounted payoff vector is

$$\tilde{V}_1(\theta) = \sum_{j=1}^K \theta_j \tilde{S}_1^j \in \mathbb{R}^N, \quad V_0(\theta) = \sum_{j=1}^K \theta_j S_0^j.$$

Finance the initial cost using the bank account. Consider the *zero-cost* discounted gain:

$$G(\theta) := \tilde{V}_1(\theta) - V_0(\theta) \mathbf{1} \in \mathbb{R}^N, \quad \mathbf{1} = (1, \dots, 1)^\top.$$

Then $G(\theta)$ is the discounted terminal P&L of a self-financing strategy with initial cost 0.

No-arbitrage as a separation statement

An arbitrage exists iff $\exists \theta$ such that $G(\theta) \geq 0$ componentwise and $G(\theta) \neq 0$. So **no arbitrage** means: $\{G(\theta) : \theta \in \mathbb{R}^K\} \cap (\mathbb{R}_+^N \setminus \{0\}) = \emptyset$.

FTAP: No Arbitrage $\Rightarrow$ Risk-Neutral Measure (Separation)

Let $\mathcal{G} := \{G(\theta) : \theta \in \mathbb{R}^K\}$. This is a linear subspace of $\mathbb{R}^N$.

Key fact (finite-dimensional *strict* separation): Since $\mathbb{R}_{++}^N$ (strictly positive orthant) is open and $\mathcal{G} \cap \mathbb{R}_+^N = \{0\}$ under no-arbitrage, there exists $\psi \in \mathbb{R}_{++}^N$ (all components > 0) such that

$$\psi^\top g = 0 \quad \text{for all } g \in \mathcal{G}.$$

Apply this to $g = G(\theta)$:

$$0 = \psi^\top (\tilde{V}_1(\theta) - V_0(\theta)1) = \psi^\top \tilde{V}_1(\theta) - V_0(\theta) \psi^\top 1 \quad \forall \theta.$$

Hence, for each asset j ,

$$S_0^j = \frac{\psi^\top \tilde{S}_1^j}{\psi^\top 1}.$$

Define a probability vector $q_i := \psi_i / (\sum_{m=1}^N \psi_m)$. Then $q_i > 0$, $\sum_i q_i = 1$, and $S_0^j = E^Q[\tilde{S}_1^j]$. Thus $Q(\omega_i) = q_i$ is an *equivalent* risk-neutral measure. ■

Completeness (one-period)

The market is *complete* if every contingent claim $H : \Omega \rightarrow \mathbb{R}$ can be replicated: $\exists \theta$ such that $V_1(\theta; \omega) = H(\omega)$ for all ω .

Linear algebra viewpoint: Let M be the $N \times K$ matrix of discounted payoffs. Replication means: solve $M\theta = H$.

Key fact

In the finite one-period model:

- **Completeness** $\iff \text{rank}(M) = N$.
- **Unique** risk-neutral measure $\iff$ completeness.

The **CRR model** is the canonical binomial tree model.

Setup:

- Time periods: $t = 0, 1, \dots, T$
- Stock can go **up** by factor $u > 1$ or **down** by factor $d < 1$
- Risk-free rate: r per period (continuously compounded)
- Bernoulli trials: $X_t \in \{0, 1\}$, $P(X_t = 1) = p$

Stock price dynamics:

$$S_t = S_0 \cdot u^{N_t} \cdot d^{t-N_t}$$

where $N_t = \sum_{s=1}^t X_s$ = number of up moves.

Key feature: Tree recombines! (up-down = down-up)

$\Rightarrow$ Only $t + 1$ nodes at time t , not 2^t .

Goal: Find probability q such that discounted stock price is a martingale.

Martingale condition:

$$S_t = e^{-r} E_q[S_{t+1} | \mathcal{F}_t]$$

Calculation: At each step:

$$S_t = e^{-r} [q \cdot S_t u + (1 - q) \cdot S_t d]$$

$$e^r = qu + (1 - q)d$$

Solving for q :

$$q = \frac{e^r - d}{u - d}$$

No-arbitrage condition: For $q \in (0, 1)$, we need $d < e^r < u$.

Intuition: Stock must not dominate or be dominated by bond.

CRR to Black-Scholes: The Convergence

Scaling: Let $\Delta t = T/n$ and set:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}, \quad e^{r\Delta t} \approx 1 + r\Delta t$$

Then: As $n \rightarrow \infty$:

$$\log(S_T/S_0) = \sum_{i=1}^n Y_i \xrightarrow{d} N\left(\left(r - \frac{\sigma^2}{2}\right) T, \sigma^2 T\right)$$

where $Y_i = \log u$ or $\log d$ with probabilities $q, 1 - q$.

This is exactly the Black-Scholes distribution!

The parallel:

CLT	CRR $\rightarrow$ BS
$\frac{S_n - n\mu}{\sigma\sqrt{n}} \rightarrow N(0, 1)$	$\log(S_T/S_0) \rightarrow N(\cdot, \cdot)$

Both are **scaling limits** of sums of i.i.d. random variables!

Step 1: Write log-return as sum.

$$\log S_T = \log S_0 + \sum_{i=1}^n Z_i$$

where $Z_i = \sigma\sqrt{\Delta t}$ (up) or $Z_i = -\sigma\sqrt{\Delta t}$ (down).

Step 2: Compute moments under Q .

$$E_q[Z_i] = q\sigma\sqrt{\Delta t} - (1 - q)\sigma\sqrt{\Delta t} = (2q - 1)\sigma\sqrt{\Delta t}$$

With $q = \frac{e^{r\Delta t} - d}{u - d} \approx \frac{1}{2} + \frac{r - \sigma^2/2}{2\sigma} \sqrt{\Delta t}$:

$$E_q[Z_i] \approx \left(r - \frac{\sigma^2}{2}\right) \Delta t$$

Step 3: Apply CLT.

$$\sum_{i=1}^n Z_i \xrightarrow{d} N\left(\left(r - \frac{\sigma^2}{2}\right) T, \sigma^2 T\right)$$

Conclusion: $S_T = S_0 e^{(r - \sigma^2/2)T + \sigma W_T}$ under Q . ■

Part V

Option Pricing Models

Geometric Brownian Motion (GBM):

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Parameters:

- S_t : Stock price at time t
- μ : Drift (expected return)
- σ : Volatility (standard deviation of returns)
- W_t : Standard Brownian motion

Key assumptions:

- Continuous trading, no transaction costs
- No dividends
- Constant volatility σ and risk-free rate r
- No arbitrage

Solving GBM: Applying Itô's Lemma

Goal: Find S_t explicitly.

Strategy: Apply Itô's lemma to $f(S) = \log S$.

Itô's lemma: For $dS = \mu S dt + \sigma S dW$:

$$d(\log S) = f'(S) dS + \frac{1}{2} f''(S) (dS)^2$$

Compute derivatives:

- $f'(S) = 1/S$
- $f''(S) = -1/S^2$
- $(dS)^2 = \sigma^2 S^2 dt$ (using $dW^2 = dt$)

Substitute:

$$\begin{aligned} d(\log S) &= \frac{1}{S} (\mu S dt + \sigma S dW) - \frac{1}{2S^2} \sigma^2 S^2 dt \\ &= \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW \end{aligned}$$

From previous slide:

$$d(\log S_t) = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t$$

Integrate from 0 to t :

$$\log S_t - \log S_0 = \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t$$

Exponentiate:

$$S_t = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right]$$

Distribution: $\log(S_t/S_0) \sim N \left((\mu - \sigma^2/2)t, \sigma^2 t \right)$

$\Rightarrow S_t$ is *log-normally* distributed.

Black–Scholes: Replication and the PDE

Let $V(t, S)$ be the time- t price of a derivative written on S_t . Assume S_t follows GBM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

Apply Itô to $V(t, S_t)$:

$$dV = (V_t + \mu S V_S + \frac{1}{2} \sigma^2 S^2 V_{SS}) dt + \sigma S V_S dW.$$

Delta-hedged portfolio: hold 1 derivative and short Δ shares: $\Pi := V - \Delta S$. Choose $\Delta := V_S$ to cancel the dW term. This makes $d\Pi$ locally riskless:

$$d\Pi = (V_t + \frac{1}{2} \sigma^2 S^2 V_{SS}) dt.$$

No-arbitrage: a locally riskless portfolio must earn the risk-free rate:

$$d\Pi = r\Pi dt = r(V - S V_S) dt.$$

Black–Scholes PDE

$$V_t + \frac{1}{2} \sigma^2 S^2 V_{SS} + r S V_S - r V = 0.$$

Black–Scholes PDE: Terminal Conditions

For a European call with strike K and maturity T :

$$V(T, S) = (S - K)^+.$$

For a European put:

$$V(T, S) = (K - S)^+.$$

Key message

The PDE depends on r and σ but **not** on the real-world drift μ . That is the mathematical expression of “risk-neutral pricing”.

The PDE derivation is a replication/no-arbitrage argument. Equivalent derivations use Girsanov + Feynman–Kac.

Risk-Neutral Valuation: Where the Expectation Comes From

The PDE implies that the discounted derivative price is a martingale under a suitable measure Q .

Under Q the stock drift becomes r :

$$dS_t = rS_t dt + \sigma S_t dW_t^Q, \quad \text{so } e^{-rt} S_t \text{ is a } Q\text{-martingale.}$$

Risk-neutral pricing for European payoff $g(S_T)$

$$V(t, S_t) = E^Q \left[e^{-r(T-t)} g(S_T) \mid \mathcal{F}_t \right].$$

Interpretation

Pricing reduces to (i) specifying the risk-neutral dynamics and (ii) computing a discounted expectation.

Deriving the Closed Form: Computing $E^Q[(S_T - K)^+]$

Under Q ,

$$S_T = S_0 \exp\left(\left(r - \frac{1}{2}\sigma^2\right)T + \sigma\sqrt{T}Z\right), \quad Z \sim N(0, 1).$$

Let $g(S_T) = (S_T - K)^+$. Then

$$C_0 = e^{-rT} E^Q[(S_T - K)^+] = e^{-rT} \left(E^Q[S_T 1_{\{S_T > K\}}] - K Q(S_T > K) \right).$$

Step 1: $Q(S_T > K) = \Phi(d_2)$, where $d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}$.

Step 2: One can show the truncated lognormal identity $E^Q[S_T 1_{\{S_T > K\}}] = S_0 e^{rT} \Phi(d_1)$, with $d_1 = d_2 + \sigma\sqrt{T}$.

Therefore:

$$C_0 = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2).$$



European call option price:

$$C = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}$$

and $\Phi(\cdot)$ is the standard normal CDF.

Put option (via put-call parity):

$$P = Ke^{-rT} \Phi(-d_2) - S_0 \Phi(-d_1)$$

Theorem: For European options with same strike K and maturity T :

$$C - P = S_0 - Ke^{-rT}$$

Proof: Consider two portfolios at $t = 0$:

Portfolio A: 1 call + Ke^{-rT} cash

Portfolio B: 1 put + 1 share of stock

At maturity T :

- If $S_T > K$: A pays $(S_T - K) + K = S_T$; B pays $0 + S_T = S_T$
- If $S_T \leq K$: A pays $0 + K = K$; B pays $(K - S_T) + S_T = K$

Same payoffs $\Rightarrow$ same prices: $C + Ke^{-rT} = P + S_0$ ■

Use: Check for arbitrage, compute put from call (or vice versa).

Stopping Times (Formal)

Filtered probability space: $(\Omega, \mathcal{F}, \mathbb{Q}, (\mathcal{F}_t)_{t \in \mathcal{T}})$ where $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$ (information increases over time).

Stopping time

A map $\tau : \Omega \rightarrow \mathcal{T} \cup \{\infty\}$ is a **stopping time** if

$$\{\tau \leq t\} \in \mathcal{F}_t \quad \forall t \in \mathcal{T}.$$

Discrete time ($\mathcal{T} = \{0, 1, \dots, T\}$): equivalent characterisation

$$\{\tau = t\} \in \mathcal{F}_t \quad \forall t.$$

Interpretation: at time t you can decide whether $\tau \leq t$ using only information in $\mathcal{F}_t$.

Examples:

- $\tau \equiv T$ is a stopping time.
- **First hitting time** $\tau_B = \inf\{t : S_t \geq B\}$ is a stopping time (proved next).
- **Last exit time** $\sup\{t \leq T : S_t \leq B\}$ is typically *not* a stopping time.

Lemma: First Hitting Times Are Stopping Times

Let $(S_t)_{t \geq 0}$ be adapted to $(\mathcal{F}_t)$ and fix a level $B \in \mathbb{R}$. Define the first hitting time

$$\tau_B := \inf\{t \geq 0 : S_t \geq B\} \quad (\inf \emptyset := \infty).$$

Claim

τ_B is a stopping time.

Proof (continuous time): For any $t \geq 0$,

$$\{\tau_B \leq t\} = \left\{ \sup_{0 \leq s \leq t} S_s \geq B \right\}.$$

If S has right-continuous paths (e.g. continuous), then

$$\left\{ \sup_{0 \leq s \leq t} S_s \geq B \right\} = \bigcup_{q \in \mathbb{Q} \cap [0, t]} \{S_q \geq B\}.$$

Each $\{S_q \geq B\} \in \mathcal{F}_q \subseteq \mathcal{F}_t$ by adaptedness, hence the union is in $\mathcal{F}_t$. So $\{\tau_B \leq t\} \in \mathcal{F}_t$ for all t . ■

American Options: Optimal Stopping (Formal)

Let $(S_t)_{t \in [0, T]}$ be adapted to the market filtration $(\mathcal{F}_t)$ and let $\beta_t = e^{-rt}$. For payoff $g(S_t)$, define discounted payoff $Y_t := \beta_t g(S_t)$.

American option price

The time- t value is the Snell envelope:

$$V_t = \beta_t^{-1} \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{[t, T]}} \mathbb{E}^Q[Y_\tau \mid \mathcal{F}_t] = \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{[t, T]}} \mathbb{E}^Q \left[e^{-r(\tau-t)} g(S_\tau) \mid \mathcal{F}_t \right].$$

Key point: Pricing an American option is an *optimal stopping* problem.

Payoff functions:

- **American call:** $g(s) = (s - K)^+$
- **American put:** $g(s) = (K - s)^+$

American Call on Non-Dividend Stock

For a non-dividend paying stock with $r > 0$, an American call should **never be exercised early**. Thus:

$$C^{\text{American}} = C^{\text{European}}$$

Proof idea: Show early exercise is suboptimal.

Suppose at time $t < T$ you consider exercising when $S_t > K$.

- **Exercise now:** Receive $S_t - K$
- **Hold:** European call is worth $C_t \geq S_t - Ke^{-r(T-t)} > S_t - K$

Since $C_t > S_t - K$ always (time value is positive), never exercise early! ■

When early exercise matters

American **puts** and American calls on **dividend-paying** stocks may optimally be exercised early. American puts are strictly more valuable than European puts.

Bellman's Principle via the Snell Envelope (Discrete Time)

Work on $(\Omega, \mathcal{F}, \mathbb{Q}, (\mathcal{F}_t)_{t=0}^T)$. Let $(\beta_t)_{t=0}^T$ be the discount factor process (e.g. $\beta_t = (1+r)^{-t}$ in a constant-rate model). Let $(G_t)_{t=0}^T$ be an adapted payoff process (e.g. $G_t = (K - S_t)^+$ for an American put).

Value process (optimal stopping)

For each t , define

$$V_t := \operatorname{ess\,sup}_{\tau \in \mathcal{T}_t} \mathbb{E}^{\mathbb{Q}} \left[\frac{\beta_\tau}{\beta_t} G_\tau \mid \mathcal{F}_t \right],$$

where $\mathcal{T}_t$ is the set of stopping times with values in $\{t, t+1, \dots, T\}$.

Bellman recursion / dynamic programming

$V_T = G_T$ and for $t < T$,

$$V_t = \max \left\{ G_t, \mathbb{E}^{\mathbb{Q}} \left[\frac{\beta_{t+1}}{\beta_t} V_{t+1} \mid \mathcal{F}_t \right] \right\} \quad \text{a.s.}$$

Proof sketch of recursion:

- For any $\tau \in \mathcal{T}_t$, either $\tau = t$ or $\tau \geq t + 1$.
- If $\tau = t$, the value equals G_t .
- If $\tau \geq t + 1$, apply tower property:

$$\mathbb{E}^{\mathbb{Q}} \left[\frac{\beta_{\tau}}{\beta_t} G_{\tau} \mid \mathcal{F}_t \right] = \mathbb{E}^{\mathbb{Q}} \left[\frac{\beta_{t+1}}{\beta_t} \mathbb{E}^{\mathbb{Q}} \left[\frac{\beta_{\tau}}{\beta_{t+1}} G_{\tau} \mid \mathcal{F}_{t+1} \right] \mid \mathcal{F}_t \right] \leq \mathbb{E}^{\mathbb{Q}} \left[\frac{\beta_{t+1}}{\beta_t} V_{t+1} \mid \mathcal{F}_t \right].$$

Taking the essential supremum over τ gives the recursion.

Optimal stopping time

Define

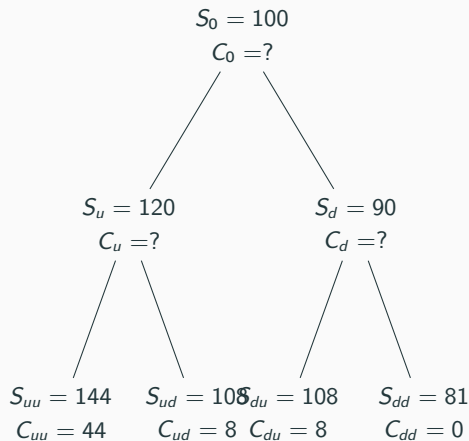
$$\tau^* := \inf \{ t \in \{0, \dots, T\} : V_t = G_t \}.$$

Then τ^* is a stopping time and it is optimal: $V_0 = \mathbb{E}^{\mathbb{Q}}[\beta_{\tau^*} G_{\tau^*}]$. Moreover, $(\beta_t V_t)$ is a supermartingale and is a martingale on $t \leq \tau^*$.

European Call: Binomial Tree Example

Setup: $S_0 = 100$, $K = 100$, $r = 5\%$, $u = 1.2$, $d = 0.9$, $T = 2$ periods

Risk-neutral probability: $q = \frac{1+r-d}{u-d} = \frac{1.05-0.9}{1.2-0.9} = 0.5$

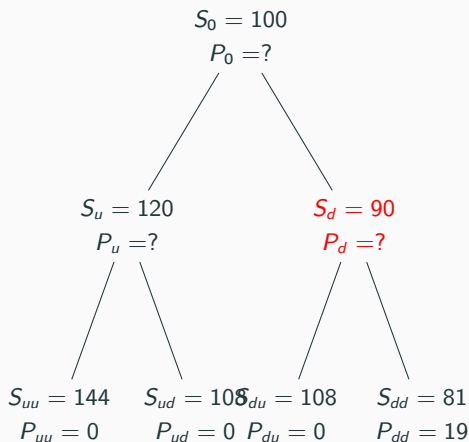


Backward induction:

$$\bullet C_u = \frac{0.5 \cdot 44 + 0.5 \cdot 8}{1.05} = 24.76, \quad C_d = \frac{0.5 \cdot 8 + 0.5 \cdot 0}{1.05} = 3.81$$

American Put: Binomial Tree Example (Corrected)

Setup: $S_0 = 100$, $K = 100$, gross rate $R = 1.05$, $u = 1.2$, $d = 0.9$, $T = 2$, $q = 0.5$.



Backward induction with early exercise:

- P_u : Continue = $\frac{0.5 \cdot 0 + 0.5 \cdot 0}{1.05} = 0$, Exercise = 0 $\Rightarrow P_u = 0$.
- P_d : Continue = $\frac{0.5 \cdot 0 + 0.5 \cdot 19}{1.05} = 9.05$, **Exercise = 10** $\Rightarrow P_d = 10$.
- P_0 : Continue = $\frac{0.5 \cdot 0 + 0.5 \cdot 10}{1.05} = 4.76$, Exercise = 0 $\Rightarrow P_0 = 4.76$.

Question: How to choose u and d to match market volatility σ ?

Cox–Ross–Rubinstein (CRR) Parametrisation

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}$$

Risk-neutral probability: $q = \frac{e^{r\Delta t} - d}{u - d}$

Jarrow–Rudd (JR) Parametrisation

$$u = e^{(r - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t}}, \quad d = e^{(r - \sigma^2/2)\Delta t - \sigma\sqrt{\Delta t}}$$

Risk-neutral probability: $q = \frac{1}{2}$

Comparison:

	CRR	JR
Recombining tree	Yes ($ud = 1$)	Yes
Risk-neutral q	Variable	Always 0.5
Centered lattice	No	Yes (drift-adjusted)

Leisen–Reimer (LR): Designed for faster convergence to Black–Scholes.

- Uses Peizer–Pratt inversion to match $\Phi(d_1)$ and $\Phi(d_2)$ directly
- Achieves $O(1/n^2)$ convergence (vs $O(1/n)$ for CRR)

Tian: Matches third moment (skewness).

$$u = \frac{M}{2} \left(V + 1 + \sqrt{V^2 + 2V - 3} \right), \quad d = \frac{M}{2} \left(V + 1 - \sqrt{V^2 + 2V - 3} \right)$$

where $M = e^{r\Delta t}$, $V = e^{\sigma^2 \Delta t}$.

Trinomial trees: Three branches (up, middle, down).

- More degrees of freedom $\Rightarrow$ can match more moments
- Natural for mean-reverting processes (interest rate models)

Practical note: For vanilla options, all methods converge to Black–Scholes. Choice matters mainly for speed and stability.

Part VI

Stochastic Calculus Essentials

Definition: Standard Brownian Motion

A stochastic process $W = (W_t)_{t \geq 0}$ is a *standard Brownian motion* if:

1. $W_0 = 0$ almost surely
2. W has **independent increments**: for $0 \leq s < t$, $W_t - W_s \perp \mathcal{F}_s$
3. W has **stationary increments**: $W_t - W_s \sim N(0, t - s)$
4. W has **continuous paths** almost surely

Key properties:

- $E[W_t] = 0$ and $\text{Var}(W_t) = t$
- $\text{Cov}(W_s, W_t) = \min(s, t)$
- Paths are continuous but **nowhere differentiable**!

Itô's Lemma: The Chain Rule for SDEs

Setup: Let X_t satisfy the SDE: $dX_t = \mu_t dt + \sigma_t dW_t$

Theorem (Itô's Lemma)

For $f(t, x) \in C^{1,2}$:

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dX_t)^2$$

Using $(dW_t)^2 = dt$, $(dt)^2 = 0$, $dt \cdot dW_t = 0$:

$$df = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dW_t$$

Key difference from ordinary calculus: The $\frac{1}{2} \sigma^2 f_{xx}$ term!

Itô's Lemma: Example 1 — $f(W_t) = W_t^2$

Goal: Find $d(W_t^2)$.

Setup: $X_t = W_t$, so $dX_t = dW_t$ (i.e., $\mu = 0$, $\sigma = 1$).

Let $f(x) = x^2$. Then $f'(x) = 2x$, $f''(x) = 2$.

Apply Itô:

$$\begin{aligned}d(W_t^2) &= f'(W_t) dW_t + \frac{1}{2} f''(W_t) (dW_t)^2 \\&= 2W_t dW_t + \frac{1}{2} \cdot 2 \cdot dt = 2W_t dW_t + dt\end{aligned}$$

Integrated form:

$$W_t^2 = \int_0^t 2W_s dW_s + t$$

Rearranging: $\int_0^t W_s dW_s = \frac{1}{2}(W_t^2 - t)$

Note: This differs from $\int x dx = x^2/2$ by the $-t/2$ term!

Itô's Lemma: Example 2 — $f(W_t) = e^{W_t}$

Goal: Find $d(e^{W_t})$.

Setup: $X_t = W_t$, $f(x) = e^x$. Then $f'(x) = f''(x) = e^x$.

Apply Itô:

$$d(e^{W_t}) = e^{W_t} dW_t + \frac{1}{2} e^{W_t} dt$$

In integral form:

$$e^{W_t} = 1 + \int_0^t e^{W_s} dW_s + \frac{1}{2} \int_0^t e^{W_s} ds$$

Observation: e^{W_t} is NOT a martingale!

Expected value: $E[e^{W_t}] = e^{t/2} \neq 1$

But: $e^{W_t - t/2}$ IS a martingale (exponential martingale).

Itô's Lemma: Example 3 — $\sin(S_t)$ for GBM

Goal: If $dS_t = \mu S_t dt + \sigma S_t dW_t$, find $d(\sin S_t)$.

Setup: $f(x) = \sin x$. Then $f'(x) = \cos x$, $f''(x) = -\sin x$.

Apply Itô:

$$\begin{aligned}d(\sin S_t) &= \cos(S_t) dS_t + \frac{1}{2}(-\sin S_t) (dS_t)^2 \\&= \cos(S_t)(\mu S_t dt + \sigma S_t dW_t) - \frac{1}{2} \sin(S_t) \cdot \sigma^2 S_t^2 dt\end{aligned}$$

Simplify:

$$d(\sin S_t) = \left(\mu S_t \cos S_t - \frac{\sigma^2 S_t^2}{2} \sin S_t \right) dt + \sigma S_t \cos S_t dW_t$$

Note: The exam question with $dS_t = \mu S_t dt + \sigma S_t dW_t$ follows exactly this pattern!

Itô's Lemma: Example 4 — Product Rule

Goal: Find $d(X_t Y_t)$ where both are Itô processes.

Setup: $dX_t = \mu^X dt + \sigma^X dW_t$, $dY_t = \mu^Y dt + \sigma^Y dW_t$.

Itô product rule:

$$d(XY) = X dY + Y dX + dX \cdot dY$$

The “extra” term: $dX \cdot dY = \sigma^X \sigma^Y dt$

Full formula:

$$d(X_t Y_t) = X_t(\mu^Y dt + \sigma^Y dW_t) + Y_t(\mu^X dt + \sigma^X dW_t) + \sigma^X \sigma^Y dt$$

Compare to ordinary calculus: $d(xy) = x dy + y dx$

The stochastic “covariance” term $dX \cdot dY$ is new!

Girsanov's Theorem: Changing Measures

Problem: Under P , stock grows at rate $\mu \neq r$.

Solution: Change to measure Q where stock grows at rate r .

Girsanov's Theorem

Let $(\theta_t)_{t \in [0, T]}$ be adapted and satisfy Novikov's condition. Define Q via the Radon–Nikodym derivative:

$$\frac{dQ}{dP} = \exp \left(- \int_0^T \theta_s dW_s - \frac{1}{2} \int_0^T \theta_s^2 ds \right)$$

Then under Q , the process $\tilde{W}_t = W_t + \int_0^t \theta_s ds$ is a standard Brownian motion.

For Black–Scholes: $\theta = (\mu - r)/\sigma$ (market price of risk).

Connection to Radon–Nikodym

Girsanov's theorem is fundamentally about measure change. The exponential martingale $\frac{dQ}{dP}$ is exactly the Radon–Nikodym derivative between the risk-neutral measure Q and the physical measure P . This derivative “re-weights” probabilities so that the discounted asset price becomes a Q -martingale.

Definition: An adapted process (M_t) is a *martingale* if:

1. $E[|M_t|] < \infty$ for all t
2. $E[M_t | \mathcal{F}_s] = M_s$ for all $s \leq t$

Interpretation: Best prediction of future value is current value.

Examples:

- Standard Brownian motion W_t
- $W_t^2 - t$ (surprisingly!)
- Discounted stock price under risk-neutral measure

Non-examples:

- e^{W_t} (this is a submartingale: $E[e^{W_t} | \mathcal{F}_s] \geq e^{W_s}$)
- Stock price under real-world measure (if $\mu > r$)

Part VII

Numerical Methods for SDEs

Why Numerical Methods for SDEs?

Problem: Most SDEs don't have closed-form solutions!

Exceptions with explicit solutions:

- **GBM:** $dS_t = \mu S_t dt + \sigma S_t dW_t \Rightarrow S_t = S_0 e^{(\mu - \sigma^2/2)t + \sigma W_t}$
- **Ornstein–Uhlenbeck:** $dX_t = -\alpha X_t dt + \sigma dW_t$

No closed form:

- **CIR:** $dr_t = \alpha(\theta - r_t) dt + \sigma \sqrt{r_t} dW_t$ (used in interest rate models)
- **CEV:** $dS_t = \mu S_t dt + \sigma S_t^\beta dW_t$ (constant elasticity of variance)
- **Heston:** Stochastic volatility model with correlated Brownian motions

Need: Systematic methods to approximate solutions numerically.

Motivation: The Picard–Lindelöf Theorem for ODEs

Consider the ODE initial value problem:

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0$$

Picard–Lindelöf Theorem

If f is Lipschitz continuous in y uniformly in t , then there exists a unique local solution.

Proof idea (Picard iteration):

1. Rewrite as integral equation: $y(t) = y_0 + \int_{t_0}^t f(s, y(s)) ds$
2. Define sequence: $y^{(0)}(t) = y_0, \quad y^{(n+1)}(t) = y_0 + \int_{t_0}^t f(s, y^{(n)}(s)) ds$
3. Show $(y^{(n)})$ is Cauchy $\Rightarrow$ converges to unique fixed point

Connection to Euler: Euler's method approximates the integral:

$$y(t_{n+1}) \approx y(t_n) + f(t_n, y(t_n)) \cdot \delta t$$

This is a *one-step* version of Picard iteration!

From ODEs to SDEs: The Key Differences

$$\text{ODE: } dy = f(t, y) dt \quad \text{vs} \quad \text{SDE: } dX = a(X) dt + b(X) dW$$

Why SDEs need special treatment:

- Brownian motion W_t is **nowhere differentiable** (a.s.)
- Paths have **unbounded variation**: $\sum |W_{t_{i+1}} - W_{t_i}| \rightarrow \infty$
- But **bounded quadratic variation**: $\sum (W_{t_{i+1}} - W_{t_i})^2 \rightarrow t$

Consequence: Ordinary calculus fails! Need Itô calculus.

SDE integral form:

$$X_t = X_0 + \int_0^t a(X_s) ds + \int_0^t b(X_s) dW_s$$

Numerical approximation:

$$X_{n+1} \approx X_n + a(X_n) \cdot \delta t + b(X_n) \cdot \Delta W_n$$

where $\Delta W_n = W_{t_{n+1}} - W_{t_n} \sim N(0, \delta t)$.

Convergence Orders: Formal Definitions

Let X_T be the true solution and $X_T^{(\delta t)}$ the numerical approximation with step size δt .

Strong Convergence Order

A numerical scheme has **strong order** γ if there exists $C > 0$ such that:

$$\mathbb{E}[|X_T - X_T^{(\delta t)}|] \leq C \cdot (\delta t)^\gamma$$

This measures **pathwise accuracy**: how close is each simulated path to the true path?

Weak Convergence Order

A numerical scheme has **weak order** β if for all smooth g with polynomial growth:

$$|\mathbb{E}[g(X_T)] - \mathbb{E}[g(X_T^{(\delta t)})]| \leq C_g \cdot (\delta t)^\beta$$

This measures **distributional accuracy**: how close are the distributions?

Convergence Orders: Intuition and Implications

Key insight: Weak convergence is easier to achieve than strong convergence!

Why?

- Strong: Must match *each path* closely
- Weak: Only need to match *statistics* (expectations)

Practical implications:

Option type	Convergence needed	Scheme
European vanilla	Weak	Euler–Maruyama OK
Asian (path average)	Strong	Milstein preferred
Barrier (path max/min)	Strong	Higher-order schemes

Computational cost: To reduce error by factor k :

- Strong order γ : need $k^{1/\gamma}$ more steps
- Example: Strong order 0.5 $\Rightarrow 4\times$ steps to halve error
- Example: Strong order 1.0 $\Rightarrow 2\times$ steps to halve error

Euler–Maruyama Scheme

SDE: $dX_t = a(X_t) dt + b(X_t) dW_t, \quad X_0 = x$

Euler–Maruyama discretisation:

$$X_{n+1} = X_n + a(X_n) \cdot \delta t + b(X_n) \cdot \Delta W_n$$

where:

- $\delta t = T/N$ (time step)
- $\Delta W_n = W_{t_{n+1}} - W_{t_n} \sim N(0, \delta t)$

Convergence:

- **Strong order:** 0.5 (pathwise approximation)
- **Weak order:** 1.0 (distribution approximation)

Rule of thumb: To halve the strong error, quarter the step size!

Milstein scheme:

$$X_{n+1} = X_n + a(X_n)\delta t + b(X_n)\Delta W_n + \frac{1}{2}b(X_n)b'(X_n)[(\Delta W_n)^2 - \delta t]$$

Where does the extra term come from?

Itô-Taylor expansion of $b(X_s)$ gives a double stochastic integral:

$$\int_{t_n}^{t_{n+1}} \int_{t_n}^s dW_u dW_s = \frac{1}{2}[(\Delta W_n)^2 - \delta t]$$

Convergence:

- **Strong order:** 1.0 (twice as good as Euler–Maruyama!)
- **Weak order:** 1.0 (same as Euler–Maruyama)

Use Milstein when: You need accurate sample paths (path-dependent options).

Given SDE:

$$dX_t = \cos(X_t) dt + \sqrt{X_t} dW_t, \quad X_0 = 7$$

Identify: $a(x) = \cos(x)$, $b(x) = \sqrt{x}$, $b'(x) = \frac{1}{2\sqrt{x}}$

Milstein scheme:

$$X_{n+1} = X_n + \cos(X_n)\delta t + \sqrt{X_n}\Delta W_n + \frac{1}{2}\sqrt{X_n} \cdot \frac{1}{2\sqrt{X_n}}[(\Delta W_n)^2 - \delta t]$$

Simplify:

$$X_{n+1} = X_n + \cos(X_n)\delta t + \sqrt{X_n}\Delta W_n + \frac{1}{4}[(\Delta W_n)^2 - \delta t]$$

Pseudocode:

1. Initialise $X_0 = 7$, choose δt
2. For $n = 0, 1, \dots, N - 1$: generate $\Delta W_n \sim N(0, \delta t)$, update X_{n+1}

Strong vs. Weak Convergence

Strong convergence of order γ :

$$\mathbb{E}[|X_T - X_T^{\delta t}|] \leq C \cdot (\delta t)^\gamma$$

$\Rightarrow$ Paths are close to true paths.

Weak convergence of order γ :

$$|\mathbb{E}[g(X_T)] - \mathbb{E}[g(X_T^{\delta t})]| \leq C \cdot (\delta t)^\gamma$$

$\Rightarrow$ Distributions are close.

When to use which?

- **European options:** Weak convergence sufficient (payoff depends only on X_T)
- **Path-dependent options:** Need strong convergence (Asian, barrier, lookback)

Summary: Euler–Maruyama is fine for European options; use Milstein for path-dependent.

1. Stochastic Runge-Kutta:

- Generalises deterministic RK methods
- Various orders possible; implementation tricky

2. Implicit (Backward) Euler: $X_{n+1} = X_n + a(X_{n+1})\delta t + b(X_n)\Delta W_n$

- Better stability for stiff SDEs
- Requires solving nonlinear equation at each step

3. Predictor-Corrector:

- Use Euler to “predict”, correct with Milstein-type step

4. Exact simulation (when possible):

- GBM: $S_t = S_0 e^{(r-\sigma^2/2)t + \sigma W_t}$ (no discretisation error!)
- OU process, CIR (via transforms)

Convergence Rate Comparison

Scheme	Strong	Weak	Cost	Best Use
Euler–Maruyama	0.5	1.0	$O(N)$	European options
Milstein	1.0	1.0	$O(N)$	Path-dependent
Stoch. RK (order 1.5)	1.5	2.0	$O(N)$	High accuracy
Implicit Euler	0.5	1.0	$O(N^2)$	Stiff SDEs
Predictor-Corrector	1.0	1.0	$O(N)$	Balanced
Exact (GBM)	∞	∞	$O(1)$	GBM/OU/CIR

Key observations:

- Strong order γ : error $\sim (\delta t)^\gamma$; Weak order γ : distributional error $\sim (\delta t)^\gamma$
- Milstein doubles strong order vs. Euler, but needs $b'(x)$
- For pricing European options, weak order matters most
- When exact simulation is possible, always use it!

Theorem (Strong Convergence)

Assume a, b are globally Lipschitz: $|a(x) - a(y)| + |b(x) - b(y)| \leq L|x - y|$.

Then Euler-Maruyama satisfies: $E[|X_T - X_T^{(n)}|] \leq C\sqrt{\delta t}$

If additionally $b \in C^1$ with b' Lipschitz, Milstein achieves: $E[|X_T - X_T^{(n)}|] \leq C\delta t$

Theorem (Weak Convergence)

Under smoothness conditions on a, b and test function g :

Both Euler-Maruyama and Milstein achieve weak order 1:

$$|E[g(X_T)] - E[g(X_T^{(n)})]| \leq C\delta t$$

Practical implication: For Monte Carlo pricing of European options, Euler-Maruyama is often sufficient—weak convergence is what matters!

Problem: MC error $\sim 1/\sqrt{N}$ (slow convergence)

1. Antithetic variates:

- For each path with ΔW , also simulate path with $-\Delta W$
- Exploits negative correlation to reduce variance

2. Control variates:

- Use known expectation of correlated quantity
- E.g., use stock price to control option price estimate

3. Importance sampling:

- Change measure to sample rare events more often
- Useful for deep OTM options, tail risk

4. Quasi-Monte Carlo:

- Use low-discrepancy sequences instead of pseudorandom
- Can achieve $O(1/N)$ convergence (vs $O(1/\sqrt{N})$)

Part VIII

Interest Rate Models

Two different objects called r

- In **equity/Black–Scholes**: r is a *constant* risk-free rate.
- In **interest-rate models**: $r(t)$ is the *short rate process*.

When both appear, we will write r for constant rates and $r(t)$ (or r_t) for the short rate.

The Term Structure of Interest Rates

Zero-coupon bond: Pays \$1 at maturity T .

Price at time t : $P(t, T)$

Yield curve: Graph of yields vs. maturity.

Continuously compounded yield:

$$y(t, T) = -\frac{\ln P(t, T)}{T - t}$$

Forward rate: Rate locked in today for future borrowing.

Instantaneous forward rate:

$$f(t, T) = -\frac{\partial \ln P(t, T)}{\partial T}$$

Short rate: $r(t) = f(t, t) =$ instantaneous borrowing rate.

Short Rate Models: Overview

Idea: Model $r(t)$ directly, derive bond prices.

Popular models:

Vasicek: $dr = a(b - r) dt + \sigma dW$

- Mean-reverting to level b
- Can go negative (problem!)

CIR: $dr = a(b - r) dt + \sigma\sqrt{r} dW$

- Mean-reverting
- Always positive (if $2ab > \sigma^2$)

Hull–White: $dr = (\theta(t) - ar) dt + \sigma dW$

- Time-dependent $\theta(t)$
- Can fit initial yield curve exactly

Definition: Affine Term Structure

A model has an *affine term structure* if:

$$P(t, T) = e^{A(t, T) - B(t, T) \cdot r(t)}$$

for deterministic functions A and B .

Why useful?

- Bond prices have closed-form solutions
- Greeks can be computed analytically
- Calibration is tractable

Sufficient condition: If drift and volatility are affine in r :

$$\mu(t, r) = \alpha(t)r + \beta(t), \quad \sigma(t, r)^2 = \gamma(t)r + \delta(t)$$

then the model admits an affine term structure. Vasicek and CIR both satisfy this!

What was LIBOR?

- London Interbank Offered Rate—the benchmark for \$350 trillion+ in contracts
- Banks submitted rates at which they *claimed* they could borrow
- Used for mortgages, derivatives, corporate loans worldwide

Why was it discontinued?

- **Manipulation scandal (2012):** Banks colluded to fix rates
- **Thin markets:** Actual interbank lending had dried up
- LIBOR was based on “expert judgment,” not transactions

Timeline:

- 2017: FCA announced LIBOR would cease after 2021
- 2021: Most LIBOR settings discontinued
- 2023: USD LIBOR (most tenors) finally ended

LIBOR Replacements: Risk-Free Rates (RFRs)

Currency	Old Rate	New RFR	Administrator
USD	USD LIBOR	SOFR	NY Fed
GBP	GBP LIBOR	SONIA	Bank of England
EUR	EURIBOR/EONIA	€STR	ECB
CHF	CHF LIBOR	SARON	SIX
JPY	JPY LIBOR	TONA	Bank of Japan

Key differences:

- RFRs are **transaction-based**, not survey-based
- RFRs are **overnight rates** (no term structure built in)
- RFRs are **nearly risk-free** (secured/central bank rates)

Challenges:

- Legacy contract transition (“tough legacy” problem)
- Building term rates from overnight rates (compounding in arrears)

Ho-Lee Model (1986)

$$dr(t) = \theta(t) dt + \sigma(t) dW(t)$$

where $\theta(t)$ is chosen to fit the initial yield curve.

Distribution of $r(t)$:

$$r(t) \sim \mathcal{N} \left(r(0) + \int_0^t \theta(u) du, \int_0^t \sigma(u)^2 du \right)$$

Pros:

- Analytical bond pricing formulas
- Fits initial curve exactly via $\theta(t)$
- Simple to implement

Cons:

- No mean reversion ($\alpha = 0$)
- Rates can go arbitrarily negative

Hull-White Model (1990)

$$dr(t) = (\theta(t) - \alpha r(t)) dt + \sigma(t) dW(t)$$

where $\alpha > 0$ is the mean reversion speed.

Solution via integrating factor:

Applying Itô to $e^{\alpha t} r(t)$ gives:

$$r(t) = r(s)e^{-\alpha(t-s)} + \int_s^t \theta(u)e^{-\alpha(t-u)} du + \int_s^t \sigma(u)e^{-\alpha(t-u)} dW(u)$$

Autocorrelation of short rate (constant σ case):

$$\text{Corr}(r(s), r(t)) = \sqrt{\frac{e^{2\alpha s} - 1}{e^{2\alpha t} - 1}}, \quad t > s > 0$$

Higher $\alpha \Rightarrow$ lower autocorrelation $\Rightarrow$ faster mean reversion.

Zero-Coupon Bond Pricing in Hull-White (Notation Fixed)

In the Hull-White model under Q :

$$dr(t) = (\theta(t) - \alpha r(t)) dt + \sigma(t) dW_t^Q.$$

Affine form

Bond prices have the form

$$P(t, T) = \exp(A(t, T) - D(t, T) r(t)), \quad D(t, T) = \frac{1 - e^{-\alpha(T-t)}}{\alpha}.$$

About $A(t)$

$A(t, T)$ is deterministic and is chosen (via θ) to fit the initial curve $P(0, T)$ exactly. In practice one works with A directly rather than writing θ explicitly.

Special case (Ho-Lee): When $\alpha = 0$, we have $D(t, T) = T - t$.

Fixed-for-floating IRS

Fix payment dates $T_1 < \dots < T_n$ and accrual factors τ_i . A **payer IRS** (pay fixed, receive floating) with notional N and fixed rate K exchanges:

- Fixed leg: $N \tau_i K$ at each T_i
- Floating leg: $N \tau_i L(T_{i-1}, T_i)$ at each T_i

(Replace L by compounded RFR in the post-LIBOR world.)

Define the swap annuity (PVBP): $A(t) := \sum_{i=1}^n \tau_i P(t, T_i)$.

Under standard assumptions, the PV of the floating leg at time $t \leq T_0$ is:

$$PV_{\text{float}}(t) = N(P(t, T_0) - P(t, T_n)).$$

The fixed leg PV is $NKA(t)$, so payer swap PV is:

$$PV_{\text{swap}}^{\text{payer}}(t) = N(P(t, T_0) - P(t, T_n) - KA(t)).$$

Par Swap Rate and Why It Matters

Par (forward) swap rate: the fixed rate $S(t)$ that makes the swap worth zero at time t :

$$0 = P(t, T_0) - P(t, T_n) - S(t) A(t).$$

Hence

$$S(t) = \frac{P(t, T_0) - P(t, T_n)}{A(t)}.$$

Key reparameterisation

Many IR options (swaptions, CMS products) are naturally options on the *swap rate* $S(t)$, with $A(t)$ acting as a natural numéraire/weight (the swap annuity).

European swaption

A **European swaption** with expiry T_0 is an option to enter an IRS at T_0 with fixed rate K .

Types:

- **Payer swaption:** right to *pay fixed* K and receive floating.
- **Receiver swaption:** right to *receive fixed* K and pay floating.
- **Bermudan:** exercise allowed on multiple dates (covered later).

Payoff at expiry T_0 (physical settlement): Using $S(T_0)$ and $A(T_0)$,

$$\text{Payoff}_{\text{payer}}(T_0) = N A(T_0) (S(T_0) - K)^+, \quad \text{Payoff}_{\text{receiver}}(T_0) = N A(T_0) (K - S(T_0))^+.$$

Settlement conventions

Swaptions can be **physically settled** (enter the swap) or **cash-settled** (exchange PV at expiry).

Why Use Swaptions? Examples

Example 1 (hedging future issuance): A firm plans to issue floating-rate debt in 6 months and fears rates may rise.

- Buy a **payer swaption** expiring in 6 months.
- If rates rise, the swaption is exercised and locks a fixed rate via the swap.

Example 2 (callable/putable products): Callable bonds, mortgage prepayment risk, Bermudan-style optionality often embed swaption-like exposure.

- Dealers hedge these books using **co-terminal swaptions** and related instruments.

Big picture

Swaptions are the *workhorse* liquid instruments for trading and hedging **interest-rate volatility**.

European Swaptions: Black (1976) Pricing Formula

Market quotes European swaptions via **Black implied volatility** on the forward swap rate.

Let $S_0 := S(0)$ be the forward swap rate and $A_0 := A(0)$ the swap annuity.

Black-76 payer swaption price

Assume under the *annuity measure* that $S(t)$ is lognormal with volatility σ_{Black} . Then the payer swaption price at $t = 0$ is:

$$\text{PS}(0) = N A_0 (S_0 \Phi(d_1) - K \Phi(d_2)),$$
$$d_1 = \frac{\ln(S_0/K) + \frac{1}{2} \sigma_{\text{Black}}^2 T_0}{\sigma_{\text{Black}} \sqrt{T_0}}, \quad d_2 = d_1 - \sigma_{\text{Black}} \sqrt{T_0}.$$

Receiver swaption: $\text{RS}(0) = N A_0 (K \Phi(-d_2) - S_0 \Phi(-d_1)).$

Why Swaptions Are Calibration Instruments

Calibration has (at least) two layers:

- **Level (yield curve):** fit $P(0, T)$ using liquid curve instruments (OIS/IRS, bonds).
- **Volatility/correlation:** fit option-implied dynamics using caps/floors and **swaptions**.

Why swaptions?

- Highly liquid across an expiry $\times$ tenor grid (the “swaption vol matrix”).
- Directly encode market prices of **interest-rate volatility** (via Black implied vols).
- Key risk drivers for callable bonds, Bermudan swaptions, CMS products.
- Identify model parameters such as $\sigma(t)$ (vol term structure) and mean reversion α .

Practical takeaway

A one-factor model can match the curve exactly and fit ATM swaptions reasonably, but cannot reproduce full smiles/skews without extra factors or stochastic volatility.

Swaptions in One-Factor Short-Rate Models: Jamshidian's Trick

At expiry T_0 , a payer swaption payoff can be written as an option on a *portfolio of zero-coupon bonds*:

$$\text{Payoff}_{\text{payer}}(T_0) = N \left(1 - P(T_0, T_n) - K \sum_{i=1}^n \tau_i P(T_0, T_i) \right)^+.$$

Jamshidian's trick (idea)

In a one-factor model, each $P(T_0, T_i)$ is a monotone function of the single state variable $r(T_0)$. Hence the expression inside $(\cdot)^+$ crosses zero at a unique r^* .

Define the **bond strikes**: $K_i := P(T_0, T_i; r^*)$ for $i = 1, \dots, n$. Then the swaption payoff equals a sum of bond put payoffs, and therefore

$$\text{PS}(0) = N \left(\sum_{i=1}^n K \tau_i \text{PutBond}(0; T_0, T_i, K_i) + \text{PutBond}(0; T_0, T_n, K_n) \right),$$

where $\text{PutBond}(0; T_0, T_i, K_i)$ is a European put on the ZCB $P(\cdot, T_i)$ exercised at T_0 .

Why this matters

In Hull–White, **bond options have closed forms**, so swaption prices become tractable.

Parametrisation of $\sigma(t)$:

- **Constant:** $\sigma(t) = \sigma$ — simple but cannot fit swaption smile
- **Piecewise constant:** $\sigma(t) = \sigma_i$ for $t \in (T_{i-1}, T_i]$ — industry standard

Strategy 1: Sequential (expiry-by-expiry) calibration

- Choose a piecewise-constant $\sigma(t) = \sigma_i$ on $(T_{i-1}, T_i]$.
- Calibrate σ_0 to match a chosen set of swaptions with expiry T_0 .
- Calibrate σ_1 to match swaptions with expiry T_1 (given σ_0), etc.
- **Pros:** Low-dimensional steps, often fast and stable.
- **Cons:** Can create a “spiky” $\sigma(t)$ and overfit local quotes.

Strategy 2: Least Squares (Levenberg-Marquardt)

$$\min_{\vec{\sigma}} \sum_{i=1}^m |V_{\text{model}}(S_i; \vec{\sigma}) - V_{\text{market}}(S_i)|^2$$

- **Pros:** Smoother $\sigma(t)$, fits all instruments reasonably
- **Cons:** Computationally heavier, no exact fit

Swaption volatility matrix: Rows = expiries, Columns = tenors

For each expiry, multiple tenors are quoted but only one σ_i available!

Approach 1: Use all instruments

- Least squares over entire vol matrix
- Good for general-purpose model

Approach 2: Hedging instruments (for Bermudans)

- Calibrate to **co-terminal swaptions**
- Example: 10NC1 Bermudan $\Rightarrow$ calibrate to 10Y/10Y, 11Y/9Y, ..., 19Y/1Y
- Ensures hedging instruments are priced correctly

Mean reversion α :

- Calibrate to correlation-dependent instruments (Bermudans)
- Or fix a priori based on historical observations (common: $\alpha \approx 0.01$ – 0.05)

Pricing Bermudan Swaptions (Notation + Measure Fixed)

Bermudan: exercise at dates $T_0 < T_1 < \dots < T_{n-1}$.

At each T_i , holder chooses:

- **Exercise:** $\text{Ex}(T_i)$ (intrinsic value / European value-at-exercise depending on convention)
- **Continue:** $\text{Cont}(T_i)$

$$\text{Berm}(T_i) = \max\{\text{Ex}(T_i), \text{Cont}(T_i)\}.$$

Backward induction under $\mathbb{Q}$ (short-rate model):

- Terminal: $\text{Berm}(T_{n-1}) = \text{Ex}(T_{n-1})$.
- For $i < n - 1$:

$$\text{Cont}(T_i) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left(- \int_{T_i}^{T_{i+1}} r(u) du \right) \text{Berm}(T_{i+1}) \mid \mathcal{F}_{T_i} \right].$$

In lattice/grid implementations one often writes the discount factor as $P(T_i, T_{i+1})$.

Implementation: Longstaff–Schwartz, 1D grids, or PDE in one-factor models.

Suggested References (Mathematical Parts)

- **Coherent risk measures:** Artzner, Delbaen, Eber & Heath (1999)
- **Risk-neutral pricing / FTAP:** Harrison & Pliska (1981); finite-state proof via separating hyperplane/Farkas lemma
- **Binomial model:** Cox–Ross–Rubinstein (1979)
- **Black–Scholes:** Black & Scholes (1973), Merton (1973)
- **Swaptions:** Black (1976) for market quoting; Jamshidian (1989) for one-factor decomposition
- **Short-rate models:** Vasicek (1977), Hull–White (1990)

Recommended textbooks:

- J.C. Hull, *Options, Futures and Other Derivatives*
- D. Brigo & F. Mercurio, *Interest Rate Models—Theory and Practice*
- T. Björk, *Arbitrage Theory in Continuous Time*
- S. Shreve, *Stochastic Calculus for Finance I & II*

Thank You!

Questions?

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